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Methods and devices for controlling the temperature of a surface

Abstract

Disclosed herein are systems and methods for a method for controlling the temperature of a surface. In some embodiments, the method includes applying a variable input to a heat pump (e.g., a Peltier) in thermal communication with the surface. The variable input may vary about a voltage and may include a powered period and an unpowered period. In some embodiments, the method includes measuring a temperature of an element in thermal communication with the surface, determining, based on the temperature of the element, whether a temperature of the surface is within a safe temperature range for a user in thermal communication with the surface, in response to determining the surface is not within the safe temperature range, adjusting the voltage of the variable input to the heat pump, and in response to determining the surface is within the safe temperature range, continuing operation of the variable input at the voltage.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application is a Divisional of U.S. patent application Ser. No. 17/956,710, filed on Sep. 29, 2022, which claims the benefit of U.S. Provisional Application No. 63/407,577, filed Sep. 16, 2022, the entirety of which is incorporated by reference herein.

FIELD

(1) This disclosure generally relates to a device for controlling the temperature of a surface in thermal communication with a user. More specifically, this disclosure relates to a thermal adjustment device and methods for operating same.

BRIEF SUMMARY

(2) Disclosed herein are systems and methods for controlling the temperature of a surface.

(3) One implementation of the present disclosure is a method for controlling the temperature of a surface. In some embodiments, the method includes applying a variable input to a heat pump (e.g., a Peltier) in thermal communication with the surface. The variable input may vary about a voltage and may include a powered period and an unpowered period. In some embodiments, the method includes measuring a temperature of an element in thermal communication with the surface, determining, based on the temperature of the element, whether a temperature of the surface is within a safe temperature range for a user in thermal communication with the surface, in response to determining the surface is not within the safe temperature range, adjusting the voltage of the variable input to the heat pump (e.g., Peltier), and in response to determining the surface is within the safe temperature range, continuing operation of the variable input at the voltage.

(4) In some embodiments, the safe temperature range for the user is between 10 degrees Centigrade (in some embodiments, 15 degrees) and 49 degrees Centigrade (in some embodiments, 58 degrees or 50 degrees). In some embodiments, a duty cycle of the variable input is at least 10%. In some embodiments, a duration of the powered period is greater than a duration of the unpowered period. In some embodiments, the voltage of the variable input is between 1.7 Volts and 4.0 Volts. In some embodiments, the voltage of the variable input is within a safe operating range for the heat pump. In some embodiments, applying the variable input to the heat pump includes operating the heat pump in two modes. Each mode of the two modes may be associated with a different safe temperature range. In some embodiments, a mode of the two modes includes two stages. Each stage of the two stages may be associated with a different duration. In some embodiments, adjusting the voltage of the variable input includes reducing an amount of power provided to the heat pump (e.g., Peltier). In some embodiments, the method includes adjusting operation of a fan based on the temperature of the element.

(5) Another implementation of the present disclosure is a lanyard to receive a thermal adjustment device. The lanyard may include a body formed of a first material having a first durometer value and a second material having a second durometer value. The lanyard may include an arm coupled to the body via a truss. In some embodiments, the truss is coupled to the body at a portion of the body formed of the second material. In some embodiments, the second durometer value is greater than the first durometer value.

(6) In some embodiments, the first durometer value is between Shore A 30 and Shore A 45. In some embodiments, the second durometer value is between Shore A 50 and Shore A 70. In some embodiments, the body includes a pocket to receive a thermal adjustment device. In some embodiments, the body reduces impacts on the thermal adjustment device. In some embodiments, the body includes an aperture to allow a heating/cooling surface of the thermal adjustment device to contact a user. In some embodiments, a vertical axis of the body subtends an angle of 0-35 degrees relative to a vertical axis of a user. In some embodiments, the body includes a contact portion configured to contact a user. In some embodiments, the contact portion includes a coefficient of friction.

(7) Another implementation of the present disclosure is a thermal adjustment device. In some

embodiments, the thermal adjustment device includes a housing including an air intake and/or a thermal heat pump positioned within the housing and configured to receive air from the air intake. In some embodiments, the air intake includes a first aperture and a second aperture that is offset from the first aperture.

(8) In some embodiments, the first aperture includes a slit and the second aperture includes a hole.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 illustrates an exemplary method for controlling the temperature of a surface, in accordance with an embodiment.

(2) FIG. 2 illustrates another exemplary method for controlling the temperature of a surface, in accordance with an embodiment.

(3) FIGS. 3A-L are graphs illustrating exemplary temperature profiles of a thermal adjustment device in operation, in accordance with an embodiment.

(4) FIG. 4 illustrates a block diagram of a thermal adjustment device, in accordance with an embodiment.

(5) FIGS. 5A and 5B illustrate an exemplary thermal adjustment device and lanyard, in accordance with an embodiment.

(6) FIGS. 5C-5E illustrate various perspectives of an exemplary thermal adjustment device inserted into a lanyard, in accordance with an embodiment.

(7) FIG. 5F illustrates an exemplary thermal adjustment device coupled to a lanyard, in accordance with an embodiment.

(8) FIG. 5G illustrate an exemplary lanyard for a thermal adjustment device, in accordance with an embodiment.

(9) FIG. 5H illustrates an exemplary thermal adjustment device, in accordance with an embodiment.

(10) FIG. 5I illustrates an exemplary protective feature of an intake of a thermal adjustment device, in accordance with an embodiment.

(11) FIG. 6A-6D illustrates an exemplary thermal adjustment device as worn by a user, in accordance with an embodiment.

(12) FIG. 7 illustrates an exemplary thermal adjustment device and corresponding features, in accordance with an embodiment.

(13) FIGS. 8A-8C illustrates an exemplary method of manufacturing an exemplary lanyard of a thermal adjustment device, in accordance with an embodiment.

DETAILED DESCRIPTION

(14) In the following description of embodiments, reference is made to the accompanying drawings which form a part hereof, and in which it is shown by way of illustration specific embodiments which can be practiced. It is to be understood that other embodiments can be used and structural changes can be made without departing from the scope of the disclosed embodiments.

(15) Referring generally to the FIGURES, described herein are systems and methods for controlling the temperature of a surface. Speaking generally, disclosed herein is a thermal adjustment device and methods for operating the thermal adjustment device. In brief, the thermal adjustment device may include a thermoelectric heat pump that may be controlled to heat and/or cool a user. For example, a user may wear the thermal adjustment device in contact with their skin and the thermal adjustment device may operate to cool the user (e.g., by transferring heat from the user's skin to the environment, etc.). The user may use the thermal adjustment device to provide a comfortable personal temperature (e.g., reduce overheating, help with hot-flashes, etc.).

Additionally or alternatively, the user may use the thermal adjustment device to provide localized

heating/cooling. For example, in a building lacking central air-conditioning, a user may wear the thermal adjustment device to stay cool while working.

(16) The thermal adjustment device of the present disclosure may offer many benefits over existing systems. As an example, traditional thermal adjustment devices may allow for ingress of objects to the device, thereby creating a hazardous situation. Traditional thermal adjustment devices may include a fan to facilitate heat transfer from the thermal adjustment device. However, because most thermal adjustment devices are positioned about a user's head (e.g., around a neck of the user, etc.), the fan of traditional thermal adjustment devices may interact with a user's hair. For example, a traditional thermal adjustment device may include a fan that may allow for the ingress of a user's hair, thereby causing discomfort and/or danger to the user. However, the thermal adjustment device of the present disclosure may include a baffle at an air intake of the fan to prevent ingress of objects such as hair.

(17) Moreover, traditional thermal adjustment devices may lack active safety monitoring. For example, a traditional thermal adjustment device may rely on operating a thermoelectric heat pump within an input range to ensure user safety (e.g., safe operating temperatures, etc.). However, a temperature of the thermoelectric heat pump may vary greatly depending on a thermal load experienced by the thermoelectric heat pump. For example, a thermoelectric heat pump (e.g., in a heating scenario) may get much hotter when coupled to a small thermal load as compared to a large thermal load (e.g., holding driving power constant, etc.). To continue the example, because traditional thermal adjustment device may lack safety monitoring, they may potentially injure users (e.g., if operated without a thermal load and then applied to a user's skin, etc.). However, the thermal adjustment device of the present disclosure may solve this problem by actively monitoring a temperature associated with the thermoelectric heat pump to ensure a safe temperature range. Moreover, the thermal adjustment device may provide a comfortable personal temperature to a user. For example, the thermal adjustment device may operate a thermal heat pump using a first voltage (e.g., a maximum voltage possible for the thermal heat pump, etc.) for a first time period (e.g., to accelerate a speed of heating/cooling a thermal mass in communication with the thermal heat pump, etc.). Traditional thermal adjustment devices may not be able to achieve similar heating/cooling capacity.

(18) Referring now to FIG. 1, method **100** for controlling the temperature of a surface is shown, according to an exemplary embodiment. In various embodiments, method **100** is performed by a thermal adjustment device. The thermal adjustment device may include a thermoelectric heat pump that may heat and/or cool a user in thermal communication with the thermoelectric heat pump. For example, a user may wear the thermal adjustment device around their neck such that the thermal adjustment device contacts the skin of the user (see, for example, FIGS. 6A-6C) and the thermal adjustment device may remove heat from the user's skin (e.g., by dissipating the heat to the environment, etc.). It should be understood that while method **100** is described in relation to a heat pump, the thermal adjustment device may include any heat pump known in the art.

(19) At step **110**, the thermal adjustment device may apply a variable input to a heat pump in thermal communication with a surface. The variable input may include a time-varying input voltage. For example, the thermal adjustment device may apply 2.8V to the heat pump for 40 seconds and then may apply 2.6V to the heat pump for 180 seconds. As another example, the thermal adjustment device may apply 2.8V to the heat pump for a first time period (e.g., 4 seconds, etc.), may measure a temperature associated with the heat pump, and in response to comparing the temperature to a threshold, may apply 2.2V to the heat pump for a second time period (e.g., 20 seconds, etc.). In some embodiments, the variable input includes a voltage range associated with a safe temperature range for a user. Additionally or alternatively, the variable input may include a safe operating voltage range for the heat pump (e.g., a voltage range that falls within an input voltage specification of the heat pump, etc.). In some embodiments, the variable input includes a number of stages. For example, the thermal adjustment device may apply a variable input having a

first stage corresponding to a first input voltage and a second stage corresponding to a second input voltage, where the second input voltage changes based on a temperature feedback mechanism. To continue the example, during the second stage, the thermal adjustment device may apply a first voltage to the heat pump and may adjust the first voltage every 4 seconds in increments of 0.05V in order to achieve a setpoint temperature at a surface of the heat pump. In various embodiments, different stages correspond to different input voltages. For example, a first stage may correspond to a uniform input voltage (e.g., a steady-state 2.6V input), while a second stage may correspond to a varying input voltage (e.g., an input voltage that varies according to a sine wave (e.g., either discretely or continuously)). As another example, a first stage may correspond to a steady-state input of 2.2V and a second stage may correspond to a steady-state input of 2.6V. In various embodiments, the thermal adjustment device transitions between stages based on various conditions as described in detail below with reference to FIG. 2.

(20) In various embodiments, the thermal adjustment device controls the variable input to ensure user safety (e.g., performs safety monitoring, etc.). For example, the thermal adjustment device may control an input voltage to a heat pump based on monitoring a temperature of a surface of the heat pump in thermal communication with a user. For example, the thermal adjustment device may apply 2.8V to the heat pump for a first time period (e.g., 4 seconds, etc.), may measure a temperature associated with the heat pump, and in response to determining the temperature exceeds a threshold (e.g., a safety threshold, etc.), may apply 2.2V to the heat pump for a second time period (e.g., 20 seconds, etc.). In various embodiments, the thermal adjustment device implements one or more modes. Modes may include, for example, a heating mode and a cooling mode. Each mode may be associated with a different power level. For example, a first mode may correspond to an input voltage of 2.2V while a second mode may correspond to an input voltage of 2.6V. Each mode may include one or more stages. For example, a first mode may include three stages, where an input voltage of each stage of the three stages is within 1.7V-2.8V. To continue the example, a second mode may also include three stages, where an input voltage of each stage of the three stages is within 1.9V-3.1V.

(21) In some embodiments, the variable input includes a powered portion (e.g., a time period corresponding to a non-zero input voltage/current) and/or an unpowered portion (e.g., a time period corresponding to a substantially zero input voltage/current). For example, during the powered portion, the thermal adjustment device may apply 2.6V (e.g., with non-zero current) to the heat pump. To continue the example, during the unpowered portion the thermal adjustment device may apply 0.0V (e.g., with zero current, with minimal leakage current, etc.) to the heat pump. Taken together, the powered portion and the unpowered portion may form a duty cycle. For example, the thermal adjustment device may apply 2.2V to the heat pump for 20 second and then 0.0V to the heat pump for 10 seconds (e.g., 66.66% duty cycle). In various embodiments, different modes and/or stages include different duty cycles. For example, a first stage of a first mode may include a 100% duty cycle (e.g., a non-zero duration powered portion and a zero duration unpowered portion, a steady-state input, etc.) and a second stage of the first mode may include a 66.66% duty cycle (e.g., a 20 second powered portion and a 10 second unpowered portion, etc.). In some embodiments, the thermal adjustment device adjusts an input voltage to the heat pump between powered/unpowered cycles. For example, the thermal adjustment device may apply 2.2V to the heat pump for 20 seconds, may then apply 0.0V to the heat pump for 10 seconds, may measure a temperature associated with the heat pump, and in response to comparing the temperature to a safety threshold, may then apply 2.25V to the heat pump for 20 seconds.

(22) At step **120**, the thermal adjustment device may measure a temperature of an element in thermal communication with the surface. For example, the thermal adjustment device may measure the temperature of a surface of the heat pump. As another example, the thermal adjustment device may measure the temperature of a thermal conductor (e.g., a heatsink, etc.) coupled to a surface of the heat pump. In various embodiments, step **120** includes measuring the temperature using a

temperature sensor. The temperature sensor may include a mechanical temperature sensor and/or an electrical temperature sensor. For example, the temperature sensor may include a thermometer, a therm, a thermistor, a thermocouple, a resistance thermometer, a silicon bandgap temperature sensor, an infrared thermometer, a fiber-optic thermometer, a distributed temperature sensing (DTS) system, a pyrometer, and/or the like. In various embodiments, the element in thermal communication with the surface include a thermal conductor. For example, the element may include a metallic element (e.g., a heat pipe, etc.) coupled to the surface. As another example, the thermal adjustment device may measure a temperature of the user's skin. Additionally or alternatively, step **120** may include measuring a temperature of the surface directly.

(23) At step **130**, the thermal adjustment device may determine whether a temperature of the surface is within a safe temperature range for a user. For example, the thermal adjustment device may compare the temperature of the surface to a range to determine whether the temperature of the surface falls within the range. As another example, the thermal adjustment device may compare the temperature to a threshold. In various embodiments, the safe temperature range is between 10 degrees Centigrade (in some embodiments, 15 degrees or 10 degrees) and 49 degrees Centigrade (in some embodiments, 58 degrees or 50 degrees). For example, the thermal adjustment device may determine whether the temperature of the surface is below 49° C.

(24) Determining whether the temperature of the surface is within the safe temperature range may include determining a temperature of the surface based on the temperature of the element in thermal communication with the surface (e.g., measured in step **120**). For example, the thermal adjustment device may determine a temperature of the surface by applying a conversion factor to the temperature measured in step **120**. As another example, the thermal adjustment device may determine a temperature of the surface by applying a transfer function to the temperature measured in step **120**. In some embodiments, step **130** includes determining a relative temperature of the surface. For example, the thermal adjustment device may determine whether the temperature of the surface is above or below the safe temperature range (and by how much, etc.). In various embodiments, the user is in thermal communication with the surface. For example, the surface may be coupled to (e.g., in direct physical contact with, etc.) skin of the user.

(25) If the temperature of the surface is not within a safe temperature range (No), then method **100** may proceed with step **140**. At step **140**, the thermal adjustment device may adjust a voltage of the variable input to the heat pump. For example, the thermal adjustment device may decrease an input voltage by 0.05V. Adjusting the voltage of the variable input may include increasing the voltage or decreasing the voltage. In various embodiments, the thermal adjustment device determines the adjustment based on the temperature of the surface. For example, if the temperature is above a threshold (e.g., a setpoint, etc.), the thermal adjustment device may decrease the voltage applied to the heat pump by 0.05V. As another example, if the temperature is below a threshold, the thermal adjustment device may increase the voltage applied to the heat pump by 0.05V.

(26) In some embodiments, the thermal adjustment device determines a step size of the adjustment based on a magnitude of the difference between the temperature of the surface and a target (e.g., a setpoint, a threshold, a temperature range, etc.). For example, the thermal adjustment device may increase an input voltage by 1.0V in response to determining a 5° C. difference between the temperature of the surface and a setpoint and may increase the input voltage by 0.5V in response to determining a 3° C. difference between the temperature of the surface and a setpoint. In some embodiments, the thermal adjustment device implements a proportional-integral derivative control scheme. For example, the thermal adjustment device may apply a large voltage adjustment when a temperature of the surface is far from a setpoint and may apply a small voltage adjustment when the temperature of the surface is close to the setpoint. In some embodiments, step **140** includes depowering the heat pump. For example, the thermal adjustment device may apply 0.0V to the heat pump. If the temperature of the surface is within a safe temperature range (Yes), then method **100** may proceed with step **150**. At step **150**, the thermal adjustment device may continue operation of

the variable input. For example, the thermal adjustment device may continue providing the same input voltage to the heat pump.

(27) Referring now to FIG. 2, method **200** for determining an input for a thermoelectric heat pump is shown, according to an exemplary embodiment. In various embodiments, the thermal adjustment device performs method **200**. For example, the thermal adjustment device may implement method **200** to heat and/or cool a user. In various embodiments, the thermoelectric heat pump includes a heat pump. In brief, the thermal adjustment device may operate the heat pump in a number of stages. Each stage may correspond to a different duration and/or input characteristics. Steps **202-206** may correspond to a first stage, steps **208-218** may correspond to a second stage, and/or steps **220-228** may correspond to a third stage. In various embodiments, the thermal adjustment device implements one or more cooling modes. In some embodiments, the thermal adjustment device implements a first stage of a number of cooling modes as described in the table below.

(28) TABLE-US-00001 First stage (e.g., 40 sec) First portion (e.g., 20 sec) Second portion (e.g., 20 sec) Input to heat pump heat pump Function First 2.8 V 2.8 V Monitor temperature periodically cooling (e.g., every 10 sec) mode If temperature < threshold (e.g., 23° C.), start second stage with 2.2 V If temperature within range (e.g., 23° C. to 25° C.), start second stage with 2.4 V If ((time ≥ 40 sec) AND (temperature < threshold) (e.g., 24.5° C.)), start second stage with 2.6 V If ((time ≥ 40 sec) AND (temperature ≥ threshold) (e.g., 24.5° C.)), start second stage with 2.8 V Second 3.4 V 3.4 V Monitor temperature periodically cooling (e.g., every 10 sec) mode If temperature < threshold (e.g., 21° C.), start second stage with 2.2 V If temperature within range (e.g., 21° C. to 23° C.), start second stage with 2.5 V If ((time ≥ 40 sec) AND (temperature < threshold) (e.g., 22.5° C.)), start second stage with 2.8 V If ((time ≥ 40 sec) AND (temperature ≥ threshold) (e.g., 22.5° C.)), start second stage with 3.4 V Third 4.0 V 4.0 V Monitor temperature periodically cooling (e.g., every 10 sec) mode If temperature < 20° C., start second stage with 2.2 V If temperature within range (e.g., 20° C. to 22° C.), start second stage with 2.8 V If ((time ≥ 40 sec) AND (temperature < threshold) (e.g., 21.5° C.)), start second stage with 3.4 V If ((time ≥ 40 sec) AND (temperature ≥ threshold) (e.g., 21.5° C.)), start second stage with 4.0 V

(29) It should be understood that while the tables of the present disclosure include exemplary monitoring periods (e.g., 4 seconds, 10 seconds, etc.), any monitoring period may be used. For example, the thermal adjustment device may monitor a temperature of a surface in continuously (e.g., every 10 milliseconds, etc.). In some embodiments, the first portion (e.g., 20 seconds, etc.) facilitates providing a timely comfortable temperature to a user (e.g., quickly cooling the user, etc.). In some embodiments, the second portion (e.g., 20 seconds, etc.) facilitates providing a comfortable temperature to a user within a time period (e.g., to avoid cooling a temperature of the heat pump too quickly, etc.). It should be understood that while the tables of the present disclosure include exemplary temperature values (e.g., thresholds, ranges, setpoints, etc.), any temperature value may be used that facilitates performance of the thermal adjustment device (e.g., is within a safe temperature range for a user, etc.). In some embodiments, the thermal adjustment device implements a second stage of the number of cooling modes as described in the table below.

(30) TABLE-US-00002 Second stage (e.g., 180 sec) Stabilization Input to heat pump Function First 1.7 V~2.8 V Monitor temperature periodically (e.g., every 4 sec) cooling Target setpoint (e.g., 23° C.) mode Start duty cycle at 50% If temperature < threshold (e.g., 23° C.), decrease PWM 10%. If PWM at 10%, decrease voltage by step size (e.g., 0.15 V) If temperature ≥ threshold (e.g., 23° C.), increase PWM 10%. If PWM at 90%, increase voltage by step size (e.g., 0.15 V) Second 1.7 V~3.4 V Monitor temperature periodically (e.g., every 4 sec) cooling Target setpoint (e.g., 21° C.) mode Start duty cycle at 50% If temperature < threshold (e.g., 21° C.), decrease PWM 10%. If PWM at 10%, decrease voltage by step size (e.g., 0.15 V) If temperature ≥ threshold (e.g., 21° C.), increase PWM 10%. If PWM at 90%, increase voltage by step size (e.g., 0.15 V) Third 1.7 V~4.0 V Monitor temperature periodically (e.g., every 4 sec) cooling Target setpoint (e.g., 20° C.) mode Start duty cycle at 50% If temperature < threshold (e.g., 20° C.), decrease PWM 10%. If PWM at

10%, decrease voltage by step size (e.g., 0.15 V) If temperature $\geq$ threshold (e.g., 20° C.), increase PWM 10%. If PWM at 90%, increase voltage by step size (e.g., 0.15 V)

(31) Additionally or alternatively, the thermal adjustment device may implement the second stage of the number of cooling modes as described in the table below.

(32) TABLE-US-00003 Second stage (e.g., 180 sec) Stabilization Input to heat pump Function First 1.7 V~2.8 V Monitor temperature periodically (e.g., every 4 sec) cooling Target setpoint (e.g., 23° C.) mode If temperature $\geq$ setpoint (e.g., 23° C.), increase input by step size (e.g., 0.05 V) If temperature $<$ setpoint (e.g., 23° C.), decrease input by step size (e.g., 0.05 V) Second 1.7 V~3.4 V Monitor temperature periodically (e.g., every 4 sec) cooling Target setpoint (e.g., 21° C.) mode If temperature $\geq$ setpoint (e.g., 21° C.), increase input by step size (e.g., 0.1 V) If temperature $<$ setpoint (e.g., 21° C.), decrease input by step size (e.g., 0.1 V) Third 1.7 V~4.0 V Monitor temperature periodically (e.g., every 4 sec) cooling Target setpoint (e.g., 20° C.) mode If temperature $\geq$ setpoint (e.g., 20° C.), increase input by step size (e.g., 0.15 V) If temperature $<$ setpoint (e.g., 20° C.), decrease input by step size (e.g., 0.15 V)

(33) In some embodiments, the thermal adjustment device implements a third stage of the number of cooling modes as described in the table below.

(34) TABLE-US-00004 Third stage Waved: Powered period (e.g., 20 sec)/unpowered period (e.g., 10 sec) Input to heat pump Function First 1.7 V~2.8 V Monitor temperature periodically (e.g., every 30 sec) cooling Target setpoint (e.g., 23° C.) mode Start with same input as end of Second Stage If temperature $<$ threshold (e.g., 23° C.), decrease PWM 10%. If PWM at 10%, decrease voltage by step size (e.g., 0.15 V) If temperature $\geq$ threshold (e.g., 23° C.), increase PWM 10%. If PWM at 90%, increase voltage by step size (e.g., 0.15 V) Second 1.7 V~3.4 V Monitor temperature periodically (e.g., every 30 sec) cooling Target setpoint (e.g., 21° C.) mode Start with same input as end of Second Stage If temperature $<$ threshold (e.g., 21° C.), decrease PWM 10%. If PWM at 10%, decrease voltage by step size (e.g., 0.15 V) If temperature $\geq$ threshold (e.g., 21° C.), increase PWM 10%. If PWM at 90%, increase voltage by step size (e.g., 0.15 V) Third 1.7 V~4.0 V Monitor temperature periodically (e.g., every 30 sec) cooling Target setpoint (e.g., 20° C.) mode Start with same input as end of Second Stage If temperature $<$ threshold (e.g., 20° C.), decrease PWM 10%. If PWM at 10%, decrease voltage by step size (e.g., 0.15 V) If temperature $\geq$ threshold (e.g., 20° C.), increase PWM 10%. If PWM at 90%, increase voltage by step size (e.g., 0.15 V)

(35) Additionally or alternatively, the thermal adjustment device may implement the third stage of the number of cooling modes as described in the table below.

(36) TABLE-US-00005 Third stage Waved: Powered period (e.g., 20 sec)/unpowered period (e.g., 10 sec) Input to heat pump Function First 1.7 V~2.8 V Monitor temperature periodically (e.g., every 30 sec) cooling Target setpoint (e.g., 23° C.) mode Start with same input as end of Second Stage If temperature $<$ threshold (e.g., 23° C.), decrease voltage by step size (e.g., 0.05 V) If temperature $\geq$ threshold (e.g., 23° C.), increase voltage by step size (e.g., 0.05 V) Second 1.7 V~3.4 V Monitor temperature periodically (e.g., every 30 sec) cooling Target setpoint (e.g., 21° C.) mode Start with same input as end of Second Stage If temperature $<$ threshold (e.g., 21° C.), decrease voltage by step size (e.g., 0.1 V) If temperature $\geq$ threshold (e.g., 21° C.), increase voltage by step size (e.g., 0.1 V) Third 1.7 V~4.0 V Monitor temperature periodically (e.g., every 30 sec) cooling Target setpoint (e.g., 20° C.) mode Start with same input as end of Second Stage If temperature $<$ threshold (e.g., 20° C.), decrease voltage by step size (e.g., 0.15 V) If temperature $\geq$ threshold (e.g., 20° C.), increase voltage by step size (e.g., 0.15 V)

(37) It should be understood that while the tables of the present disclosure include exemplary time periods (e.g., a duration of a powered/unpowered period, etc.), any time period may be used that facilitates performance of the thermal adjustment device. Additionally or alternatively, the thermal adjustment device may implement a number of heating modes. In some embodiments, the thermal adjustment device implements a first stage of a number of heating modes as described in the table below.

(38) TABLE-US-00006 First stage First portion (e.g., 40 sec) Input to heat pump Function First 2.2 V Monitor temperature periodically (e.g., every 1 sec, every 4 sec) heating If temperature > threshold (e.g., 38° C.), start second mode stage with 1.75 V If time > 40 sec, start second stage with 1.75 V Second 2.8 V Monitor temperature periodically (e.g., every 1 sec, every 4 sec) heating If temperature > threshold (e.g., 39° C.), start second mode stage with 1.8 V If time > 40 sec, start second stage with 1.8 V Third 3.4 V Monitor temperature periodically (e.g., every 1 sec, every 4 sec) heating If temperature > threshold (e.g., 40° C.), start second mode stage with 1.9 V If time > 40 sec, start second stage with 1.9 V

(39) In some embodiments, the thermal adjustment device implements a second stage of a number of heating modes as described in the table below.

(40) TABLE-US-00007 Second stage Stabilization Input to heat pump Function First 1.7 V~2.2 V Monitor temperature periodically (e.g., every 1 sec, every 4 sec) heating If temperature > threshold (e.g., 46° C.), power off until mode temperature ≤ threshold (e.g., 39.5° C.) If temperature < threshold (e.g., 39.5° C.), increase voltage by step size (e.g., 0.05 V) Second 1.7 V~2.8 V Monitor temperature periodically (e.g., every 1 sec, every 4 sec) heating If temperature > threshold (e.g., 46° C.), power off until mode temperature ≤ threshold (e.g., 40.5° C.) If temperature < threshold (e.g., 40.5° C.), increase voltage by step size (e.g., 0.1 V) Third 1.7 V~3.4 V Monitor temperature periodically (e.g., every 1 sec, every 4 sec) heating If temperature > threshold (e.g., 46° C.), power off until mode temperature ≤ threshold (e.g., 41.5° C.) If temperature < threshold (e.g., 41.5° C.), increase voltage by step size (e.g., 0.15 V)

(41) Additionally or alternatively, the thermal adjustment device may implement the second stage of the number of heating modes as described in the table below.

(42) TABLE-US-00008 Second stage Stabilization Input to heat pump Function First 1.7 V~2.2 V Monitor temperature periodically (e.g., every 1 sec, every 4 sec) heating Duty cycle starts at 10% (e.g., 1.7 V at 90% & 2.2 V at 10%) mode If temperature < threshold (e.g., 39.5° C.), increase duty cycle 10% (e.g., to a maximum of 90%) If temperature ≥ threshold (e.g., 46° C.), power off until temperature 0.2° C. Second 1.7 V~2.8 V Monitor temperature periodically (e.g., every 1 sec, every 4 sec) heating Duty cycle starts at 10% (e.g., 1.7 V at 90% & 2.2 V at 10%) mode If temperature < threshold (e.g., 40.5° C.), increase duty cycle 10% (e.g., to a maximum of 90%) If temperature ≥ threshold (e.g., 46° C.), power off until temperature 0.2° C. Third 1.7 V~3.4 V Monitor temperature periodically (e.g., every 1 sec, every 4 sec) heating Duty cycle starts at 10% (e.g., 1.7 V at 90% & 2.2 V at 10%) mode If temperature < threshold (e.g., 41.5° C.), increase duty cycle 10% (e.g., to a maximum of 90%) If temperature ≥ threshold (e.g., 46° C.), power off until temperature 0.2° C.

(43) As mentioned above, the thermal adjustment device implements a number of modes. Each mode may correspond to a different heating/cooling intensity (e.g., low, medium, high, etc.). For example, a first mode may deliver a first amount of cooling (e.g., heat transfer) to a user and a second mode may deliver a second amount of cooling to a user that is greater than the first amount of cooling. In various embodiments, the thermal adjustment device achieves different levels of heating/cooling by applying different input voltages to the heat pump, thereby changing a temperature of a surface in thermal communication with the user. For example, the thermal adjustment device may cool a user a small amount by applying 1.0V to the heat pump and may cool the user a larger amount by applying 2.0V to the heat pump. Method **200** may correspond to a single mode. For example, method **200** may correspond to a first mode having a first input voltage range to achieve a first amount of heat transfer. In various embodiments, a user selects different heating/cooling modes using a user interface as described below.

(44) To implement different modes (e.g., different amounts of heat transfer, different intensities of heating/cooling, etc.), the thermal adjustment device may implement method **200** using different voltage values. For example, in a first cooling mode the thermal adjustment device may implement method **200** using a first voltage range and in a second cooling mode the thermal adjustment device may implement method **200** using a second voltage range (e.g., where voltages of the first voltage

range are less than voltages of the second voltage range, etc.). Therefore, it should be understood that while method **200** is described in relation to various input voltages, other voltages may be used (e.g., corresponding to different modes, etc.).

(45) At step **202**, the thermal adjustment device may apply a first voltage to a heat pump. For example, the thermal adjustment device may apply 2.8V to a heat pump (e.g., create an electrical potential of 2.8V between inputs of the heat pump, etc.). In some embodiments, the thermal adjustment device applies the first voltage to the heat pump for a time period. For example, in a first cooling mode (e.g., low cooling) the thermal adjustment device may apply 2.8V to the heat pump for a period of 20 seconds. As another example, in a second cooling mode (e.g., medium cooling) the thermal adjustment device may apply 3.4V to the heat pump for 20 seconds. As yet another example, in a third cooling mode (e.g., high cooling) the thermal adjustment device may apply 4.0V to the heat pump for 20 seconds.

(46) At step **204**, the thermal adjustment device may measure a temperature of an element in thermal communication with the heat pump. For example, the thermal adjustment device may measure the temperature of a surface of the heat pump. As another example, the thermal adjustment device may measure the temperature of a thermal conductor coupled to a surface of the heat pump and determine a temperature of a surface of the heat pump based on the temperature of the thermal conductor as described above with reference to FIG. 1.

(47) At step **206**, the thermal adjustment device may determine whether a first condition is met. Determining whether the first condition is met may include determining whether the temperature of the element is within a temperature range. For example, the thermal adjustment device may compare a temperature of a surface of the heat pump (e.g., determined in step **204**) to a threshold. Additionally or alternatively, determining whether the first condition is met may include determining whether the first input has been applied for a threshold time period. For example, the thermal adjustment device may determine that the first condition is met if the thermal adjustment device has operated in the first stage (e.g., steps **202-206**) for 40 seconds and a temperature of a surface of the heat pump is greater than a threshold. In some embodiments, the thermal adjustment device operates in the first stage for a time period (e.g., to allow the thermal adjustment device to build up a thermal mass, to timely provide a comfortable personal temperature to a user, etc.). In some embodiments, the time period is adjusted to change a battery life of the thermal adjustment device (e.g., the time period is shortened to increase a battery life of the thermal adjustment device, etc.).

(48) In various embodiments, the first condition includes a number of sub-conditions. For example, the first condition may be implemented as:

IF ((temperature<23° C.) OR (temperature<24.5° C. AND time≥40 seconds) OR (temperature>24.5° C. AND time≥40 seconds)) THEN (perform step 208)

(49) In some embodiments, steps **204-206** are performed periodically. For example, the thermal adjustment device may perform steps **204-206** every 4 seconds. To continue the example, the thermal adjustment device may apply the first voltage to the heat pump for 20 seconds and after the 20 seconds may monitor a temperature of the heat pump every 4 seconds to determine whether the first condition is satisfied (e.g., steps **204-206**).

(50) If the first condition is not satisfied (No), method **200** may proceed with step **202** (e.g., continue applying the first voltage to the heat pump). In various embodiments, the thermal adjustment device repeats steps **202-206** until the first condition (or one of the number of sub-conditions) is satisfied. If the first condition is satisfied (Yes), method **200** may proceed with step **208**. At step **208**, the thermal adjustment device may apply a second voltage to the heat pump. For example, the thermal adjustment device may apply 2.6V to the heat pump. In some embodiments, the thermal adjustment device applies the second voltage to the heat pump for a time period. For example, in a first cooling mode (e.g., low cooling) the thermal adjustment device may apply the second voltage to the heat pump for a period of 180 seconds.

(51) In various embodiments, the second voltage is determined based on a result of step **206**. For example, if the temperature of the surface of the heat pump is less than 23° C. at step **206**, then the thermal adjustment device may apply 2.2V to the heat pump. In some embodiments, the second voltage is the same as the first voltage. In various embodiments, the thermal adjustment device determines the second voltage according to the relationships:

IF (temperature < 23° C.).fwdarw.second voltage = 2.2V

IF (temperature < 24.5° C. AND time ≥ 40 seconds).fwdarw.second voltage = 2.6V

IF (temperature > 24.5° C. AND time ≥ 40 seconds).fwdarw.second voltage = 2.8V

(52) At step **210**, the thermal adjustment device may measure a temperature of the element. For example, the thermal adjustment device may measure the temperature of a surface of the heat pump. Additionally or alternatively, the thermal adjustment device may measure the temperature of the element using any other means described with respect to step **120** above. At step **212**, the thermal adjustment device may determine whether a second condition is met. For example, the thermal adjustment device may determine whether the second voltage has been applied for a threshold amount of time (e.g., 180 seconds, etc.). In various embodiments, the second condition is implemented as:

IF (time ≥ 180 seconds) THEN (perform step 220)

(53) If the first condition is not satisfied (No), method **200** may proceed with step **214**. If the first condition is satisfied (Yes), method **200** may proceed with step **220**. At step **214**, the thermal adjustment device may determine whether the temperature of the element is higher or lower than a threshold. In various embodiments, the threshold is a setpoint. For example, in a first cooling mode the threshold may be a setpoint of 23° C. As another example, in a second cooling mode the threshold may be a setpoint of 21° C. As yet another example, in a third cooling mode the threshold may be a setpoint of 20° C.

(54) If the temperature of the element is higher than the threshold, method **200** may proceed with step **216**. If the temperature of the element is lower than the threshold, method **200** may proceed with step **218**. In various embodiments, if the temperature of the element is equal to the threshold, then method **200** proceeds with step **210** (e.g., at the next sampling interval, etc.). At step **216**, the thermal adjustment device may decrease the second voltage applied to the heat pump. For example, the thermal adjustment device may decrease the second voltage by 0.05V. At step **218**, the thermal adjustment device may increase the second voltage applied to the heat pump. For example, the thermal adjustment device may increase the second voltage by 0.05V. In various embodiments, the thermal adjustment device repeats steps **210-218** until the second condition is met.

(55) At step **220**, the thermal adjustment device may apply a third voltage to the heat pump. For example, the thermal adjustment device may apply the same voltage applied in step **208**. To continue the example, the thermal adjustment device may apply a voltage of 2.6V in step **208**, may increase the voltage in steps 0.05V to 2.75V in steps **210-218**, and may apply 2.75V in step **220**.

(56) At step **222**, the thermal adjustment device may measure the temperature of the element. For example, the thermal adjustment device may measure the temperature of a surface of the heat pump. Additionally or alternatively, the thermal adjustment device may measure the temperature of the element using any other means described with respect to step **120** above. At step **224**, the thermal adjustment device may determine whether the temperature of the element is higher or lower than a threshold. For example, the thermal adjustment device may compare the temperature to a threshold. In various embodiments, the threshold is a setpoint. For example, in a first cooling mode the threshold may be a setpoint of 23° C. As another example, in a second cooling mode the threshold may be a setpoint of 21° C. As yet another example, in a third cooling mode the threshold may be a setpoint of 20° C. In some embodiments, the threshold of step **224** is different than the threshold of step **214**. Additionally or alternatively, the threshold of step **224** may be the same as the threshold of step **214**.

(57) If the temperature of the element is higher than the threshold, method **200** may proceed with

step **226**. If the temperature of the element is lower than the threshold, method **200** may proceed with step **228**. In various embodiments, if the temperature of the element is equal to the threshold, then method **200** proceeds with step **222** (e.g., at the next sampling interval, etc.). At step **226**, the thermal adjustment device may increase the voltage applied to the heat pump. For example, the thermal adjustment device may increase the third voltage by 0.05V. At step **228**, the thermal adjustment device may decrease the voltage applied to the heat pump. For example, the thermal adjustment device may decrease the third voltage by 0.05V. In various embodiments, steps **222-228** are performed periodically. For example, the thermal adjustment device may monitor a temperature of the element and update the third voltage every 30 seconds. However, it should be understood that other periods are possible (e.g., every second, every hour, etc.). Moreover, it should be understood that other voltage adjustments amounts (e.g., 1.0V, 1.1V, 4.0V, etc.) are possible.

(58) Referring now to FIGS. **3A-L**, a number of graphs illustrating temperature profiles of a thermal adjustment device are shown, according to an exemplary embodiment. In various embodiments, the graphs illustrate a temperature of a contact surface of the thermal adjustment device during different stages of an operating mode. FIG. **3A** illustrates temperature measurements of a surface in thermal communication with a user during three stages of a first cooling mode. In some embodiments, the surface is directly contacting the user's skin; in other embodiments, intermediate structure separates the surface and the user's skin. FIG. **3B** illustrates temperature measurements of a surface in thermal communication with a user during three stages of a second cooling mode. FIG. **3C** illustrates temperature measurements of a surface in thermal communication with a user during three stages of a third cooling mode. FIG. **3D** illustrates temperature measurements of a surface in thermal communication with a user during two stages of a first heating mode. FIG. **3E** illustrates temperature measurements of a surface in thermal communication with a user during two stages of a second heating mode. FIG. **3F** illustrates temperature measurements of a surface in thermal communication with a user during two stages of a third heating mode. As shown, the graphs include surface temperature **310** and ambient temperature **320**. Surface temperature **310** may correspond to a temperature of a surface of the thermal adjustment device. For example, surface temperature **310** may be a temperature of a surface of the thermal adjustment device that is in thermal communication with a user (e.g., in contact with a user's skin, etc.). Ambient temperature **320** may correspond to a temperature of the ambient environment. For example, ambient temperature **320** may correspond to a temperature of a room the thermal adjustment device is operating within.

(59) In various embodiments, the thermal adjustment device operates in various stages. For example, in a first cooling mode (e.g., shown in FIG. **3A**), the thermal adjustment device may operate in three stages: (i) an initialization stage, (ii) a stabilization stage, and (iii) a waved cool stage. As another example, in a first heating mode (e.g., shown in FIG. **3D**), the thermal adjustment device may operate in two stages: (i) an initialization stage and (ii) a stabilization stage. FIGS. **3A-3C** are shown to include first stage **330**, second stage **340**, and third stage **350**. First stage **330** may correspond to steps **202-206** of FIG. **2**. Second stage **340** may correspond to steps **208-218** of FIG. **2**. Third stage **350** may correspond to steps **220-228** of FIG. **2**. FIGS. **3D-3F** are shown to include first stage **360** and second stage **370**.

(60) In brief, in first stage **330** the thermal adjustment device may apply a first voltage to the heat pump for a time period (e.g., 20 second, 40 seconds, etc.) and/or until a temperature threshold is met. For example, the thermal adjustment device applies 2.8 Volts to the heat pump for 20 seconds. In some embodiments, the time period is static (e.g., 20 seconds). Additionally or alternatively, the time period is dynamic (e.g., changes based on conditions). For example, the thermal adjustment device applies 2.8 Volts to the heat pump until a temperature of a surface of the heat pump is determined to exceed a threshold. In some embodiments, the first voltage corresponds to a maximum cooling intensity. Advantageously, in first stage **330** the thermal adjustment device may operate the heat pump at a maximum cooling intensity to timely provide a comfortable personal

temperature to a user (e.g., cool the user quickly). In second stage **340**, the thermal adjustment device may apply a second voltage to the heat pump for a time period (e.g., 180 seconds, etc.) and may adjust the second voltage to achieve a temperature setpoint (e.g., threshold, etc.). In third stage **350**, the thermal adjustment device may apply a third voltage to the heat pump during a powered period (e.g., 20 seconds, etc.), may remove power from the heat pump during an unpowered period (e.g., 10 second, etc.), and may adjust the third voltage to achieve a temperature setpoint (e.g., a threshold, etc.). In various embodiments, third stage **350** corresponds to a waved cooling mode (e.g., an oscillation between two temperatures, etc.). Advantageously, by varying a temperature of the surface, the waved cooling mode may provide a comfortable personal temperature for a user (e.g., as opposed to cooling to a static temperature, etc.). In various embodiments, a waved cooling mode reduces a user's habituation to a cooling sensation (e.g., as opposed to cooling to a static temperature, etc.), thereby improving a user's subjective sense of cooling relief. In various embodiments, the temperature setpoints are associated with a mode. For example, in a first cooling mode (e.g., low) the temperature setpoint may be 23° C. In a second cooling mode (e.g., medium) the temperature setpoint may be 21° C. In a third cooling mode (e.g., high) the temperature setpoint may be 20° C.

(61) Referring still to FIGS. **3A-3C**, in first stage **330**, the thermal adjustment device may apply a first voltage to the heat pump (thereby causing a surface in thermal communication with a user to cool) for a time period (e.g., 20 seconds). At the end of the time period, the thermal adjustment device may determine that the temperature of the surface is lower than a threshold (e.g., 23° C.) and may transition to second stage **340**. In second stage **340**, the thermal adjustment device may apply a second voltage (e.g., 2.2V, etc.) to the heat pump for a time period (e.g., 180 second, etc.). In various embodiments, the second voltage is determined based on a temperature of the surface in first stage **330**. For example, in a first cooling mode the thermal adjustment device may implement the function:

IF (temperature<23° C.).fwdarw.terminate first stage and begin second stage with 2.2V

IF (temperature<24.5° C. AND time≥40 seconds).fwdarw.terminate first stage and begin second stage with 2.6V

IF (temperature≥24.5° C. and time≥40 seconds).fwdarw.terminate first stage and begin second stage with 2.8V

(62) At the end of the time period, the thermal adjustment device may transition to third stage **350**. In third stage **350**, the thermal adjustment device may apply a variable input including a powered period and an unpowered period. For example, the thermal adjustment device may apply an input having a 66% duty cycle. In some embodiments, the powered period is 20 seconds and the unpowered period is 10 seconds. However, it should be understood that other durations for the powered period and/or the unpowered period are possible. In some embodiments, the thermal adjustment device implements the function:

IF (temperature<threshold (e.g., 23° C.)).fwdarw.decrease a voltage of the variable input by step size (e.g., 0.05V)

IF (temperature>threshold (e.g., 23° C.)).fwdarw.increase a voltage of the variable input by step size (e.g., 0.05V)

(63) It should be understood that other temperature thresholds and/or voltage adjustments (e.g., steps of 1V, 2V, etc.) are possible. For example, in a second cooling mode, the thermal adjustment device may use a 21° C. threshold. As another example, in a third cooling mode, the thermal adjustment device may use a 20° C. threshold. It should be understood that, as used herein, a temperature measurement may refer to a temperature measurement associated with a heat pump. For example, the thermal adjustment device may measure a temperature of a thermal conductor (e.g., a heatsink, etc.) in thermal communication with a heat pump (e.g., Peltier, etc.). As another example, the thermal adjustment device may measure a temperature of a surface of the heat pump.

(64) Referring now specifically to FIGS. **3D-3F**, in first stage **360**, the thermal adjustment device

may apply a first voltage to the heat pump for a set time period (e.g., 20 second, 40 seconds, etc.) and/or until a temperature threshold is met. In second stage **370**, the thermal adjustment device may apply a second voltage to the heat pump and may adjust the second voltage based on monitoring a temperature of a surface of the thermal adjustment device (e.g., to achieve a temperature setpoint, to ensure a safe surface temperature for the user, etc.).

(65) In first stage **360**, the thermal adjustment device may apply a first voltage to the heat pump (thereby causing a surface in thermal communication with a user to warm) for a time period (e.g., 20 seconds). If the temperature of the surface is higher than a threshold (e.g., 38° C., etc.), the thermal adjustment device may transition to second stage **370**. Additionally or alternatively, at the end of the time period, the thermal adjustment device may transition to second stage **370**.

(66) In second stage **370**, the thermal adjustment device may apply a second voltage (e.g., 1.75V, etc.) to the heat pump. The thermal adjustment device may monitor a temperature of the surface periodically (e.g., every 4 seconds, etc.). In response to a temperature of the surface, the thermal adjustment device may perform an action (e.g., adjust an input voltage to the heat pump, etc.). In various embodiments, the thermal adjustment device implements the function.

IF (temperature<threshold (e.g., 39.5° C.)).fwdarw.increase a voltage of the variable input by step size (e.g., 0.05V)

IF (temperature≥threshold (e.g., 46° C.)).fwdarw.set a voltage of the variable input to a value (e.g., 0.0V)

(67) It should be understood that other temperature thresholds, voltage adjustments (e.g., steps of 1V, 2V, etc.), and/or variable input values (e.g., 0.01V, etc.) are possible. Moreover, it should be understood that the thresholds may be the same and/or different. For example, the thermal adjustment device may utilize a first threshold as a setpoint and a second threshold as a safe operating temperature ceiling. The thermal adjustment device may implement one or more safe temperature routines. For example, a first routine may determine whether a temperature of a surface of the thermal adjustment device is outside a first range (e.g., 15 degrees Centigrade to 49 degrees Centigrade, etc.) and a second routine may determine whether a temperature of the surface is outside a second range (e.g., 10 degrees Centigrade to 50 degrees Centigrade, etc.). In some embodiments, the one or more routines are executed by different components of the thermal adjustment device (e.g., to provide redundancy, etc.). For example, a first processor may execute a first routine and a second processor may execute a second routine. In some embodiments, the thermal adjustment device implements one or more temperature monitoring routines. For example, a first routine may determine whether a temperature of a surface of the thermal adjustment device is outside a safe temperature range for a user (e.g., 10 degrees Centigrade to 50 degrees Centigrade) and a second routine may determine whether a temperature of the surface of the thermal adjustment device is outside a temperature range associated with a current operating mode (e.g., 41.5 degrees Centigrade to 46 degrees Centigrade for a third heating mode, etc.).

(68) Referring now specifically to FIGS. 3G-3L, input voltage **380** associated with temperature **382** is shown, according to an exemplary embodiment. In various embodiments, the thermal adjustment device applies input voltage **380** to a heat pump (e.g., a Peltier, etc.) to achieve temperature **382** (given ambient temperature **384**) at a surface of the heat pump (e.g., a surface in thermal communication with a user, etc.). As shown, FIGS. 3G-3I include first stage **390**, second stage **392**, and third stage **394**. However, it should be understood that the thermal adjustment device may implement greater/fewer stages. Each stage may correspond to a different functionality. For example, first stage **390** corresponds to a cooling initialization stage and may include applying a static input to the heat pump. As another example, second stage **392** corresponds to a cooling stabilization stage and may include applying a dynamic input (e.g., an input that changes in response to one or more conditions such as a temperature measurement, etc.) to the heat pump. As another example, third stage **394** corresponds to a waved cooling stage and may include applying a variable input. In various embodiments, the variable input includes powered period **393A** and

unpowered period **393B**. Powered period **393A** may correspond to a time period (e.g., 20 seconds, etc.) during which the thermal adjustment device applies a first voltage to the heat pump. Unpowered period **393B** may correspond to a time period (e.g., 10 seconds, etc.) during which the thermal adjustment device applies a second voltage to the heat pump. In various embodiments, the first voltage is greater than the second voltage. For example, the first voltage may be 2.6 Volts and the second voltage may be 0.25 Volts. In some embodiments, the first voltage is dynamic (e.g., changes in response to one or more conditions such as a temperature measurement). For example, the thermal adjustment device applies 2.6 Volts to the heat pump, measures a temperature of a surface of the heat pump, and in response determining whether the surface corresponds to a safe operating temperature for a user (e.g., based on the temperature measurement), applies 2.4 Volts to the heat pump. In some embodiments, the second voltage is zero (or substantially close to zero). In various embodiments, each stage is associated with a different input to the heat pump (as described in greater detail with reference to FIG. 2, above). FIG. 3G corresponds to a first cooling mode (e.g., low cooling intensity). FIG. 3H corresponds to a second cooling mode (e.g., medium cooling intensity). FIG. 3I corresponds to a third cooling mode (e.g., high cooling intensity). FIG. 3J corresponds to a first heating mode (e.g., low heating intensity). FIG. 3K corresponds to a second heating mode (e.g., medium heating intensity). FIG. 3L corresponds to a third heating mode (e.g., high heating intensity).

(69) As shown, FIGS. 3J-3L include first stage **396** and second stage **398**. First stage **396** may correspond to a heating initialization stage and may include applying a static input to the heat pump. Second stage **398** may correspond to a heating stabilization stage and may include applying a dynamic input (e.g., an input that changes in response to one or more conditions such as temperature, etc.) to the heat pump. FIG. 3J corresponds to a first heating mode (e.g., low heating intensity). FIG. 3K corresponds to a second heating mode (e.g., medium heating intensity). FIG. 3L corresponds to a third heating mode (e.g., high heating intensity). The thermal adjustment device may transition between the stages described in FIGS. 3G-3L in response to one or more conditions such as time and/or temperature (e.g., a temperature measurement corresponding to a surface of a heat pump, etc.). In various embodiments, the thermal adjustment device transitions between the modes described in FIGS. 3G-3L in response to a user input (e.g., a user selecting an operating mode using a user interface of the thermal adjustment device, etc.).

(70) Referring now to FIG. 4, thermal adjustment device **400** is shown, according to an exemplary embodiment. Thermal adjustment device **400** may be used by a user to provide a comfortable temperature. For example, thermal adjustment device **400** may be used to heat and/or cool a user. In some embodiments, thermal adjustment device **400** includes processor **402**, memory **404**, heat pump **410**, temperature sensor **412**, fan **414**, user interface **416**, energy storage device **418** and/or input/output **420**. Elements of thermal adjustment device **400** may be connected with communication link **422**. In various embodiments, communication link **422** is wired. For example, communication link **422** may include a wired bus. It should be understood that the components of thermal adjustment device **400** can be connected in any suitable manner, such as via a physical bus, or wirelessly.

(71) Processor **402** may include hardware for executing instructions, such as those making up a computer program. As an example and not by way of limitation, to execute instructions, processor **402** may retrieve (or fetch) the instructions from an internal register, an internal cache, memory **404**; decode and execute them; and then write one or more results to an internal register, an internal cache, or memory **404**. Processor **402** may include one or more internal caches for data, instructions, or addresses.

(72) Memory **404** can be any suitable device that provides storage, such as an electrical, magnetic, or optical memory including a RAM, cache, hard drive, or removable storage disk. In some embodiments, memory **404** includes data **406A** and program **406B**. Data **406A** and/or program **406B** may store instructions to cause processor **402** to perform the methods disclosed herein. In

particular embodiments, memory **404** includes main memory for storing instructions (e.g., program **406B**) for processor **402** to execute or data **406A** for processor **402** to operate on. In particular embodiments, one or more memory management units (MMUs) reside between processor **402** and memory **404** and facilitate accesses to memory **404** requested by processor **402**. In particular embodiments, memory **404** includes random access memory (RAM). This disclosure contemplates any suitable RAM.

(73) Program **406B** can be stored and/or transported within any non-transitory, computer-readable storage medium for use by or in connection with an instruction execution system, apparatus, or device, such as those described above, that can fetch instructions associated with the software from the instruction execution system, apparatus, or device and execute the instructions. In the context of this disclosure, a computer-readable storage medium can be any medium, such as memory **404**, that can contain or store programming for use by or in connection with an instruction-execution system, apparatus, or device.

(74) Program **406B** can also be propagated within any transport medium for use by or in connection with an instruction-execution system, apparatus, or device, such as those described above, that can fetch instructions associated with the software from the instruction-execution system, apparatus, or device and execute the instructions. In the context of this disclosure, a transport medium can be any medium that can communicate, propagate, or transport programming for use by or in connection with an instruction-execution system, apparatus, or device. The transport readable medium can include, but is not limited to, an electronic, magnetic, optical, electromagnetic, or infrared wired or wireless propagation medium.

(75) Heat pump **410** may transfer heat from a first side of heat pump **410** to a second side of heat pump **410**. It should be understood that while heat pump **410** is described as a heat pump, it may be or include any thermoelectric heat pump. For example, heat pump **410** may include a Peltier. Heat pump **410** may include one or more components. For example, heat pump **410** may include a Peltier coupled between a heat conducting plate (e.g., a thermal mass, etc.) and a heatsink. In various embodiments, thermal adjustment device **400** operates heat pump **410** via one or more inputs. For example, thermal adjustment device **400** may apply a first voltage to an input of heat pump **410** to cause heat pump **410** to transfer heat from a first surface to a second surface (e.g., cool the first surface). To continue the previous example, thermal adjustment device **400** may apply a second voltage (e.g., having an opposite polarity as the first voltage) to an input of heat pump **410** to cause heat pump **410** to transfer heat from the second surface to the first surface (e.g., heat the second surface). In various embodiments, heat pump **410** includes one or more surfaces. For example, heat pump **410** may include a first surface on a first side of heat pump **410** configured to be positioned in thermal communication with a user's skin and a second surface on a second side of heat pump **410** configured to be positioned in thermal communication with a heatsink/radiator. In some embodiments, a surface area of the one or more surfaces is greater than traditional thermal adjustment devices. In some embodiments, the surface (e.g., a surface in thermal communication with a user) is sized to fit a user. For example, the surface may be sized to fit between a user's shoulder blades (e.g., the surface may be 2.36 inches across, etc.). As another example, the surface may be 55.5 millimeters (e.g., in width) by 60 millimeters (e.g., in length). Advantageously, by including a larger surface area, thermal adjustment device **400** may increase heat transfer with a thermal mass (e.g., a user's skin) while remaining sizable to a user (e.g., not too large to fit between the user's shoulder blades, etc.). Moreover, a surface of heat pump **410** may include a larger thermal mass than traditional thermal adjustment devices. Advantageously, by including a larger thermal mass, thermal adjustment device **400** may more effectively heat/cool a thermal mass in communication with heat pump **410** (e.g., a user's skin).

(76) In various embodiments, thermal adjustment device **400** receives information from temperature sensor **412**. Temperature sensor **412** may be and/or include any system/device configured to capture temperature information. For example, temperature sensor **412** may include a

thermometer, a therm, a thermistor, a thermocouple, a resistance thermometer, a silicon bandgap temperature sensor, an infrared thermometer, a fiber-optic thermometer, a DTS system, a pyrometer, and/or the like.

(77) Fan **414** may facilitate ambient air heat transfer for one or more components of thermal adjustment device **400**. For example, fan **414** may draw in ambient air to heat and/or cool a surface of heat pump **410**. In various embodiments, fan **414** draws in air from an ingress portion of thermal adjustment device **400** (e.g., an intake, etc.) and expels the air from an egress portion of thermal adjustment device **400** (e.g., an exhaust, etc.). Fan **414** may be or include a direct-current (DC) brushless blower. In some embodiments, fan **414** operates during a cooling mode. Additionally or alternatively, fan **414** may operate during a heating mode. In various embodiments, thermal adjustment device **400** operates fan **414** based on one or more operating parameters (e.g., a temperature of a heat pump, an ambient temperature, etc.). In various embodiments, the ingress and/or egress portions of thermal adjustment device **400** include a structure to prevent particulate matter, hair, etc. from entering thermal adjustment device **400**. For example, thermal adjustment device **400** may include an ingress portion having two layers of vents (e.g., a first horizontal layer and a second layer including a number of holes, etc.) that are offset to prevent a user's hair from ingress into thermal adjustment device **400**.

(78) In various embodiments, a user operates thermal adjustment device **400** via user interface **416**. User interface **416** may be configured to receive an input from a user (e.g., a touch input, a button input, a voice input), and a setting of thermal adjustment device **400** may be updated in response to receiving the input from the user (e.g., to change between heating and cooling modes, to change a temperature setpoint, etc.). User interface **416** may include a button, a switch, a slider, a dial, and/or the like. In various embodiments, user interface **416** includes a display element. For example, user interface **416** may include one or more light-emitting diodes (LEDs) that indicate a current heating/cooling mode of thermal adjustment device **400**. As another example, user interface **416** may include a number of LEDs to indicate a battery charge.

(79) Energy storage device **418** may store energy to power thermal adjustment device **400**. In various embodiments, energy storage device **418** includes a battery. Energy storage device **418** may be integral to thermal adjustment device **400**. Additionally or alternatively, energy storage device **418** may be positioned remotely of thermal adjustment device **400**. For example, thermal adjustment device **400** may include a power cord that connects to an external battery bank. In various embodiments, energy storage device **418** may have a greater energy capacity than traditional thermal adjustment devices. For example, energy storage device **418** may have an energy capacity sufficient to operate thermal adjustment device **400** for 4 hours (e.g., in a specific heating/cooling mode, etc.). Additionally or alternatively, thermal adjustment device **400** may facilitate higher current levels between energy storage device **418** and heat pump **410** than traditional thermal adjustment devices. Advantageously, by including a greater energy capacity/current levels, thermal adjustment device **400** may operate a heat pump using larger power inputs (e.g., voltage, current, etc.), thereby facilitating a comfortable personal temperature (e.g., because of a greater ability to quickly heat/cool heat pump **410**, etc.).

(80) Input/output **420** may include one or more interfaces. For example, thermal adjustment device **400** may include a universal serial bus (USB) type C connection to facilitate charging energy storage device **418**.

(81) Referring now generally to FIGS. 5A-5H, thermal adjustment device **500** and lanyard **600** are shown, in accordance with an embodiment. In brief, thermal adjustment device **500** may couple within lanyard **600** and be worn by a user to position a surface of thermal adjustment device **500** in thermal communication with the user's skin. For example, lanyard **600** positions thermal adjustment device **500** on a user's upper back such that a heat pump of thermal adjustment device **500** directly contacts the user's skin. Additionally or alternatively, the heat pump may indirectly contact a user (e.g., be separated by a layer of material such as an undershirt, etc.). In various

embodiments, thermal adjustment device **500** is a personal air-conditioning (AC) device.

(82) FIGS. **5A** and **5B** illustrate an exemplary thermal adjustment device **500** and lanyard **600**, in accordance with an embodiment. As shown in FIGS. **5A** and **5B**, thermal adjustment device **500** is inserted into pocket **630** of lanyard **600**. Pocket **630** may be or include a volume within lanyard **600** to receive thermal adjustment device **500**. Pocket **630** may at least partially surround thermal adjustment device **500** to secure thermal adjustment device **500** within pocket **630**. In some embodiments, a size (e.g., volume, dimensions, etc.) of pocket **630** (e.g., pocket **630** without thermal adjustment device **500** inserted, etc.) is different than a size of thermal adjustment device **500**. For example, pocket **630** may be smaller than thermal adjustment device **500** (e.g., 0.5% smaller in scale than thermal adjustment device **500**, etc.) to facilitate a secure fit of thermal adjustment device **500** within pocket **630** (e.g., via friction). In some embodiments, one or more components of thermal adjustment device **500** mechanically couple with lanyard **600** to facilitate a secure fit of thermal adjustment device **500** within pocket **630**. For example, while inserting thermal adjustment device **500** into pocket **630**, a user may stretch lanyard **600** around thermal adjustment device **500** such that controls **520** mechanically couple within a user interface portion **620**, thereby securing thermal adjustment device **500**. As another example, pocket **630** may include a raised portion (shown as lip **802** in FIG. **8A**) to facilitate a mechanical connection with surface **540**. In various embodiments, pocket **630** is shaped substantially similarly to thermal adjustment device **500**. For example, thermal adjustment device **500** may have an outer curvature and pocket **630** may include the curvature to facilitate a secure fit of thermal adjustment device **500**. Pocket **630** includes a skin contact opening (shown as opening **632**) into which thermal adjustment device **500** is positioned. When positioned in pocket **630**, lanyard **600** may provide a protective case for thermal adjustment device **500**. In some embodiments, opening **632** is larger than surface **540**. For example, opening **632** may have a radius that is 1.5 millimeters less than a radius of surface **540**. In various embodiments, a size of opening **632** is less than thermal adjustment device **500** (e.g., to prevent thermal adjustment device **500** from passing through opening **632** without user intervention such as stretching lanyard **600** around thermal adjustment device **500**). Pocket may include a number of openings (e.g., first aperture **612**, second aperture **614**, third aperture **616**, etc.) to facilitate operation of thermal adjustment device **500**. In some embodiments, the number of openings facilitate heat transfer between thermal adjustment device **500** and the ambient environment (e.g., when in a cooling mode, etc.). In some embodiments, pocket **630** is ergonomic and/or hypoallergenic, at least in portions that contact a user's skin. Pocket **630**, in some embodiments, extends further in the region of controls **520** (e.g., shown in FIG. **5H**) of thermal adjustment device **500**, thereby providing additional protection of controls **520**.

(83) As shown in FIG. **5A**, pocket **630** also includes transparent strip **618**. Strip **618** allows for operational LEDs, positioned on/integrated within thermal adjustment device **500**, to be visible to a user when thermal adjustment device **500** is positioned in pocket **630**. In various embodiments, the LEDs may be occluded from a user (e.g., not visible to the user) when off. Strip **618** may be or include a thin layer of material (e.g., silicon, etc.) that allows light from the LEDs to be viewed by a user. The LEDs may provide operational information to the user (e.g., battery level, current operating mode, etc.).

(84) FIGS. **5C-5E** illustrate various perspectives of an exemplary thermal adjustment device **500** inserted into lanyard **600**, in accordance with an embodiment. The lanyard's arms (shown as arms **640**) may include a clasp (shown as fastener **644**) opposite pocket **630** to fix the arms around a neck of a user (see FIGS. **6A-6C** below). In some embodiments, the clasp is a magnetic clasp or a ball and socket clasp (unclashed arms are illustrated in FIGS. **5F** and **5G** below). As shown in FIG. **5D**, thermal adjustment device **500** includes a heat transfer plate (shown as surface **540**) that contacts the user skin for heating and/or cooling. In various embodiments, lanyard **600** exposes the heat transfer plate of thermal adjustment device **500**. For example, lanyard **600** may insulate thermal adjustment device **500** while exposing the heat transfer plate to allow a user to position the

heat transfer plate in thermal communication with their skin (e.g., while insulating other portions of the thermal adjustment device from the user's skin). Additionally or alternatively, lanyard **600** may protect thermal adjustment device **500** from impact. For example, lanyard **600** may protect thermal adjustment device **500** from damage if dropped. In various embodiments, the heat transfer plate is constructed of a thermally conductive material (e.g., a metal, etc.).

(85) FIG. 5F illustrates an exemplary thermal adjustment device **500** coupled to lanyard **600**, in accordance with an embodiment. In the configuration illustrated in FIG. 5F, lanyard **600** is unclasp and thermal adjustment device **500** is positioned in pocket **630** of lanyard **600**. FIG. 5G illustrates lanyard **600** of FIG. 5F with thermal adjustment device **500** removed. Lanyard **600** may include body **610** coupled to arms **640** via trusses **642**. In various embodiments, body **610** includes an internal pocket to receive thermal adjustment device **500** (shown as pocket **630**). Body **610** is shown to include first aperture **612**, second aperture **614**, third aperture **616**, strip **618**, and user interface portion **620**.

(86) Advantageously, lanyard **600** may include one or more apertures (e.g., holes, shown as first aperture **612** and second aperture **614**, etc.) to facilitate airflow to the thermal adjustment device. For example, pocket **630** of lanyard **600** may include one or more holes that may allow airflow to thermal adjustment device **500** (e.g., to facilitate heat transfer with a thermal heat pump such as a heat pump, etc.). User interface portion **620** may be or include an aperture that facilitates user operation of user interface **520**. For example, user interface portion **620** may include a hole in body **610** that allows a user to interact with one or more buttons, switches, sliders, dials, and/or the like of thermal adjustment device **500**.

(87) FIG. 5H illustrates the exemplary thermal adjustment device **500** without lanyard **600**, in accordance with an embodiment. Thermal adjustment device **500** may include body **510**, intake **512**, exhaust **514**, display **516**, input/output **518**, user interface **520**, and/or fastener(s) **530**. Body **510** may be or include a case. For example, body **510** may include a plastic case that houses components of thermal adjustment device **500** (e.g., processor **402**, memory **404**, heat pump **410**, temperature sensor **412**, fan **414**, user interface **416**, energy storage device **418**, and/or input/output **420**, etc.). Intake **512** may be or include an air intake to facilitate heat transfer with one or more components of thermal adjustment device **500**. For example, intake **512** may include a number of holes to allow a fan of thermal adjustment device to draw in air (via intake **512**) and expel the air (via exhaust **514**), thereby reducing heat buildup associated with a heat pump during a cooling mode. In various embodiments, intake **512** and/or exhaust **514** include one or more features (shown as first protective feature **513A** and second protective feature **513B**) to prevent particulate ingress. For example, intake **512** may include a first and second layer of apertures (e.g., slits, holes, etc.) that are aligned to prevent a user's hair from entering into thermal adjustment device **500** (e.g., where the hair could interact with a fan of the thermal adjustment device, etc.). As another example, exhaust **514** may include a grille (as show in FIG. 5E) to prevent particulate ingress. In some embodiments, the grille improves exhaust air flow (e.g., by reducing turbulent airflow, by encouraging laminar airflow, etc.). In some embodiments, first protective feature **513A** include one or more horizontal slits (e.g., slits in body **510**). In some embodiments, second protective feature **513B** includes a surface having one or more holes. Second protective feature **513B** may be positioned between first protective feature **513A** and a fan (shown as fan **560** in FIG. 5I). In various embodiments, first protective feature **513A** and second protective feature **513B** are aligned such that particulate (e.g., hair, etc.) must pass through first protective feature **513A** and second protective feature **513B** before contacting an internal component (e.g., a fan, etc.) of thermal adjustment device **500**. In some embodiments, exhaust **514** includes one or more features to direct a direction of exhaust airflow. For example, exhaust **514** may include one or more adjustable fins to control a direction of exhaust air.

(88) Display **516** may include one or more display elements such as LEDs that may display information associated with thermal adjustment device **500** (e.g., a battery level, a cooling/heating

mode, a temperature setpoint, etc.). For example, in response to a user selecting a low cooling/heating mode, display **516** displays a first indication (e.g., illuminates a single blue/red LED, etc.). As another example, in response to a user selecting a medium cooling/heating mode, display **516** displays a second indication (e.g., illuminates two blue/red LEDs, etc.) As yet another example, in response to a user selecting a high cooling/heating mode, display **516** displays a third indication (e.g., illuminating three blue/red LEDs, etc.). In some embodiments, the LEDs include red-green-blue (RGB) LEDs. In various embodiments, display **516** includes a transparent (or semitransparent) portion of body **510** that facilitates light from one or more LEDs positioned within body **510** to be visible to a user. For example, thermal adjustment device **500** may include an array of internal LEDs and a light guide that may direct light from the LEDs to a portion of body **510** corresponding to display **516**.

(89) Input/output **518** may include an interface such as a USB type C connection to facilitate charging thermal adjustment device **500**. In various embodiments, display **516** updates based on a charging/battery status. For example, display **516** may blink a number (e.g., three, etc.) of LEDs for a time period (e.g., five seconds, etc.) to indicate that a battery is empty. In some embodiments, display **516** flashes a first LED to indicate very low charge. In some embodiments, display **516** illuminates a first LED and flashes a second LED to indicate low/medium charge. In some embodiments, display **516** illuminates a first and second LED and flashes a third LED to indicate medium/high charge. In some embodiments, display **516** illuminates a first, second, and third LED to indicate full charge. In some embodiments, after indicating full charge, display **516** may turn off (e.g., turn off the first, second, and third LEDs, etc.). User interface **520** may facilitate user operation of thermal adjustment device **500**. In some embodiments, user interface **520** includes first button **522**, second button **524**, and/or switch **526**. First button **522** may increase a heating/cooling mode (e.g., heating/cooling intensity, etc.). For example, a user may press first button **522** to increase a heating mode from low to medium. Second button **524** may decrease a heating/cooling mode (e.g., heating/cooling intensity, etc.). For example, a user may press second button **524** to decrease a heating mode from high to medium. Switch **526** may facilitate selecting different modes (e.g., heating mode, cooling mode, off, etc.). In various embodiments, switch **526** includes a number of positions (e.g., three positions, etc.). For example, a user may move switch **526** from a first position to a second position to change thermal adjustment device **500** from an off mode to a cooling mode (e.g., turn on thermal adjustment device **500**, etc.).

(90) Fastener(s) **530** may include a screw, bolt, snap fastener, magnetic fastener, and/or any other mechanical fastener. Additionally or alternatively, fastener(s) **530** may include a tongue-and-groove fastener. For example, components of thermal adjustment device **500** (e.g., portions of a case of thermal adjustment device **500**, etc.) may be fastened using a combination of screws and tongue-and-groove fasteners. Advantageously, the combination may provide increased structural strength compared with other fastening methods (e.g., a snap-fit construction, etc.). In some embodiments, fastener(s) **530** include one or more adhesives. For example, one or more components of thermal adjustment device **500** may be fastened using adhesive tape.

(91) FIG. 5I illustrates an exemplary protective feature of an intake of thermal adjustment device **500**, in accordance with an embodiment. As discussed above, intake **512** may include second protective feature **513B** to prevent a user's hair (or other particulate) from entering into thermal adjustment device and contacting fan **560**. In various embodiments, second protective feature **513B** includes a number of holes that restrict particulate ingress while allowing airflow to fan **560**. Second protective features **513B** may be offset from first protective feature **513A** (e.g., offsetting the holes from the slits, etc.), thereby creating a barrier to particulate ingress (e.g., making it difficult for hair to ingress into intake **512**, etc.).

(92) FIG. 6A-6C illustrates an exemplary thermal adjustment device **500** worn by a user, in accordance with an embodiment. In some embodiments, lanyard **600** is comprised of materials having different durability ratings. For example, pocket **630**, in some embodiments, is constructed

of silicon rubber Shore A30, but truss **642** (shown in FIG. 6A) is composed of Shore A60 durometer. The different materials may have different physical characteristics (e.g., stiffness, etc.). This configuration may advantageously provide additional structural support and flexibility, while increasing the device's resistance to impact. Multiple durometer ratings can be used. In some embodiments, the durometer ranges from 30 to 60. In some embodiments, truss **642** provides stiffness to reduce force on the user's throat when wearing.

(93) In some embodiments, openings on lanyard **600** (shown as first aperture **612** and second aperture **614**, respectively) for intake and exhaust of thermal adjustment device **500** (e.g., shown as intake **512** and exhaust **514**, respectively) are placed to align with venting (e.g., shown as exhaust **514**). A cooling mode is illustrated in FIG. 6B, with an ambient blue intake **512** and red warm exhaust **514**. In various embodiments, intake **512** and/or exhaust **514** include structural features to prevent an ingress of particulate matter and/or hair. For example, intake **512** may include a first set of horizontal slits (shown as first protective feature **513A**) and a second set of holes (shown as second protective feature **513B**) to prevent ingress of a user's hair into thermal adjustment device **500**. In various embodiments, first aperture **513A** and second aperture **513B** are offset, thereby reducing a possibility of particulate ingress. FIG. 6C illustrates an advantageous placement of thermal adjustment device **500** under a shirt and directly contacting the user's skin. In some embodiments, a silicon rubber contact patch (illustrated under the thermal skin-contact surface, shown as contact portion **650**) increases the coefficient of friction of the device and thereby reduces movement of thermal adjustment device **500** and/or lanyard **600** by contact with skin (or another surface such as an undershirt). In some embodiments, body **510** forms angle **660** with the user's upper back. In some embodiments, angle **660** ranges from 0-35 degrees. Advantageously, angle **660** may increase a contact force between thermal adjustment device **500** and/or lanyard **600** and a user's skin (or other contact surface such as an undershirt, etc.), thereby reducing movement of thermal adjustment device **500** and/or lanyard **600**.

(94) FIG. 6D illustrates exemplary trusses **642** of lanyard **600**, in accordance with an embodiment. Trusses **642** may include first dimension **643A** (e.g., a height) and second dimension **643B** (e.g., a width). First dimension **643A** may range from 10 mm-30 mm. Second dimension **643B** may range from 15 mm-30 mm. In various embodiments, trusses **642** are constructed of a material having a stiffness sufficient to prevent/reduce mechanical deformation (e.g., bending, twisting, etc.). For example, trusses **642** may reduce a deformation of arms **640** away from a user's back, thereby facilitating secure contact between a heat conducting surface of thermal adjustment device **500** and a user's skin. Advantageously, by preventing/reducing mechanical deformation, trusses **642** may broaden a contact area on a neck/shoulders of a user and thereby reduce pressure (e.g., of arms **640**, etc.) on a throat area of a user. For example, trusses **642** may change a point of deformation of arms **640** such that a load of lanyard **600** is distributed on a user's shoulders. In some embodiments, the point of deformation is 25 mm-35 mm (e.g., measured along second dimension **643B** from a point where arms **640** contact a body of lanyard **600**, etc.). In various embodiments, first portion **643C** of trusses **642** is in mechanical compression while second portion **643D** of trusses **642** is in mechanical tension. In various embodiments, trusses **642** couple to a body of lanyard **600** above a center of gravity of thermal adjustment device **500** when inserted into lanyard **600**, thereby reducing deformation of arms **640** away from a user's back.

(95) FIG. 7 illustrates an exemplary thermal adjustment device **500** and corresponding features, in accordance with an embodiment. As shown, thermal adjustment device **500** may include lanyard **600** that is configured to insulate a surface of thermal adjustment device **500** from a user's skin. As describe above, lanyard **600** may provide impact resistance (shown as impact resistance **710**) for thermal adjustment device **500**. Lanyard **600** may include one or more of fastener **644** to facilitate coupling/uncoupling thermal adjustment device **500** and/or lanyard **600** with a user (e.g., by securing lanyard **600** around a neck/shoulder region of a user, etc.). Fastener **644** may include a snap fastener, a magnetic fastener (shown as magnetic fastener **720**), a clip, and/or the like.

Lanyard **600** may be positioned at a non-zero angle relative to a vertical surface of a user's body (shown as angle **740**) such that lanyard **600** reduces movement of lanyard **600** and/or thermal adjustment device **500** relative to a surface (e.g., a user's skin, etc.). Lanyard **600** may include a contact portion (shown as non-slip portion **750**) that may increase a coefficient of friction between a user's skin and thermal adjustment device **500**/lanyard **600**, thereby facilitating a secure fit of thermal adjustment device **500** against a user's skin. Lanyard **600** may include one or more air vents (shown as first aperture **612** and second aperture **614** in FIG. 5G above) to facilitate airflow (shown as airflow **730**) to thermal adjustment device **500**.

(96) FIG. 8A-8C illustrates an exemplary method of manufacturing an exemplary lanyard **600** of thermal adjustment device **500**, in accordance with an embodiment. In the exemplary embodiment of FIGS. 8A-8C, raw material (for example, the red silicon could be Silicon Rubber Shore A60, while the white silicon could be Silicon Rubber Shore A45) placed to produce the advantageous increase in structural support and flexibility, while also increasing the device's resistance to impact. The different colors are for illustrative purposes. As shown in FIG. 8A, a number of pieces of raw material (shown as first material **810**, second material **812**, third material **820**, fourth material **822**, fifth material **824**, and sixth material **826**) may be formed/combined into first composite material **840** and second composite material **850**. In some embodiments, first composite material **840** includes Silicon Rubber Shore A 70. In some embodiments, second composite material **850** includes Silicon Rubber Shore A 45. In some embodiments, materials **810-812** are the same (e.g., Silicon Rubber Shore A 60, Silicon Rubber Shore A 70, etc.). In some embodiments, materials **820-826** are the same (e.g., Silicon Rubber Shore A 45). In various embodiments, a material having a first durometer value (e.g., stiffness, etc.) is used for a first portion of lanyard **600** and a material having a second durometer value is used for a second portion of lanyard **600**. For example, as shown in FIG. 8B, first composite material **840** may be used along a top portion of lanyard **600** to connect between trusses **642** while second composite material **850** is used for the rest of lanyard **600**, thereby providing increased rigidity along a top portion of lanyard **600** such that lanyard **600** is suitably positioned against a user's back during use. As another example, Silicon Rubber Shore A 70 is used to create first composite material **840** and Silicon Rubber Shore A 45 is used to create second composite material **850**.

(97) FIG. 8C illustrates another method of manufacturing using the same color for raw material of different strength. As shown, a number of pieces of raw material (shown as first material **810**, third material **820**, and seventh material **830**) may be positioned to form a composite component (e.g., lanyard **600**, etc.). In some embodiments, first material **810** has a first durometer value that is greater than (e.g., stiffer than, etc.) a second durometer value of third material **820** and/or seventh material **830**.

(98) Generally, as used herein, the term “substantially” is used to describe element(s) or quantitie(s) ideally having an exact quality (e.g., fixed, the same, uniformed, equal, similar, proportional), but practically having qualities functionally equivalent to the exact quality. For example, an element or quantity is described as being substantially fixed or uniformed can deviate from the fixed or uniformed value, as long as the deviation is within a tolerance of the system (e.g., accuracy requirements, etc.). As another example, two elements or quantities described as being substantially equal can be approximately equal, as long as the difference is within a tolerance that does not functionally affect a system's operation.

(99) Likewise, although some elements or quantities are described in an absolute sense without the term “substantially”, it is understood that these elements and quantities can have qualities that are functionally equivalent to the absolute descriptions. For example, in some embodiments, a ratio is described as being one. However, it is understood that the ratio can be greater or less than one, as long as the ratio is within a tolerance of the system (e.g., accuracy requirements, etc.).

(100) Although the disclosed embodiments have been fully described with reference to the accompanying drawings, it is to be noted that various changes and modifications will become

apparent to those skilled in the art. Such changes and modifications are to be understood as being included within the scope of the disclosed embodiments as defined by the appended claims.

(101) The terminology used in the description of the various described embodiments herein is for the purpose of describing particular embodiments only and is not intended to be limiting. As used in the description of the various described embodiments and the appended claims, the singular forms “a”, “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will also be understood that the term “and/or” as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. It will be further understood that the terms “includes,” “including,” “comprises,” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

Claims

1. A thermal adjustment device, comprising: a housing comprising an air intake; a cooling fan positioned within the housing; and a thermal heat pump positioned within the housing and configured to receive air from the air intake; and wherein: the air intake includes a first aperture and a second aperture that is offset from the first aperture, the first aperture is on an exterior portion of the air intake; and the second aperture is between the first aperture and the cooling fan.
 2. The thermal adjustment device of claim 1, wherein the first aperture comprises a horizontal slit and the second aperture comprises a hole, an area of the hole smaller than an area of the horizontal slit.
 3. The thermal adjustment device of claim 1, further comprising an exhaust configured to expel air from the thermal heat pump.
 4. The thermal adjustment device of claim 3, wherein the exhaust comprises a grille.
 5. The thermal adjustment device of claim 3, wherein the exhaust comprises an adjustable fin configured to control a direction of the expelled air.
 6. The thermal adjustment device of claim 1, wherein the cooling fan is configured to receive air from the air intake.
 7. The thermal adjustment device of claim 6, wherein the cooling fan is configured to reduce a temperature of the thermal heat pump.
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